VIRTUAL PERFORMANCE SOLUTION

Installation Guide

version
2023

VIRTUAL PERFORMANCE SOLUTION 2023

Installation Guide

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Hardware and Software Requirements

Platform Support

Platform Definitions

Platform types mentioned in this chapter are detailed below.

linux-libc217-x86-64

This platform is targeted at our 64-bit Linux x86_64 users.

CPU	Any x86-64 processor that supports the AVX2 and SSE4.1 extensions (such as the AMD Epyc and Intel Xeon)
Executable format	64-bit
Interpreter	Bash interpreter
Libc	Glibc-2.17
Intel FORTRAN version	19.1.0.166
MPI availability	Intel MPI Version 2018 Update 3 (default) Intel MPI Version 2019 Update 11 (embedded)
License Daemon	flexlm 11.18.2

windows-msvc142 -x86-64

This platform is targeted at our 64-bit Windows users.

CPU	Any x86-64 processor that supports the AVX2 and SSE4.1 extensions (such as the AMD Epyc and Intel Xeon)
Executable format	64-bit
Visual Studio version	2019
Intel FORTRAN version	19.1.0.166
MPI availability	Intel MPI Version 2018 Update 3 (default) Intel MPI Version 2019 Update 9 (embedded)
License Daemon	flexlm 11.18.2

Validated Platforms

Validated platforms are platforms that were tested with ESI Group's software through a set of manual or automated tests and that can properly run ESI Group's software.

	linux-libc217-x86-64	windows-msvc142 -x86-64
Solver	• CentOS 7.9 • SUSE SLES 15 SP2 • RedHat Enterprise Linux 8.6	• Windows 11 version 22H2 • Windows 10 1809 • Windows Server 2019 • Windows Server 2016

Supported Platforms

Supported platforms are platforms relying on the platform vendor 's compatibility rules and that should run properly ESI Group's software.

	linux-libc217-x86-64	windows-msvc142 -x86-64
Solver	• RedHat Enterprise Linux 7 • RedHat Enterprise Linux 8 • CentOS 7 • SUSE SLES 12 SP5 • SUSE SLES 15 SP3	• Windows 10 1809 and newer

Not Supported Platforms

Not supported platforms are platforms relying on the platform vendor's compatibility rules and that may not run properly ESI Group's software.

	linux-libc217-x86-64	windows-msvc142 -x86-64
Solver	• CentOS6	• Windows 8 and older • Windows Server 2008 and 2012

Two-year Plan for Platform Support

This information is based on ESI Group’s current knowledge of its product's supported platform evolution. It is intended for information purpose only.

		2024	**2025**
Linux	RedHat Entreprise Linux 7 / CentOS 7	**End of support**	
	RedHat Entreprise Linux 8 / CentOS 8	Supported	Supported
	SUSE SLES 12 SP5	**End of support**	
	SUSE SLES 15 SP3	Supported	Supported
	SUSE SLES 15 SP4	Supported	Supported
Windows	Windows 11	Supported	Supported
	Windows 10 from 1809	Supported	Supported
	Windows Server 2019	Supported	Supported

MPI Support

Validated MPI Version

Validated MPI versions are MPI versions that were tested with ESI Group's software through a set of manual or automated tests and that can properly run ESI Group's software.

Platform	MPI	Interconnect
linux-libc217-x86-64	• Intel MPI 2018.3 (default) • Intel MPI 2019.11 (embedded)	• Mellanox Infiniband • Intel Onmi-Path
windows-msvc142 -x86-64	• Intel MPI 2018.3 (default) • Intel MPI 2019.9 (embedded)	• Ethernet

Supported MPI Versions

Supported MPI versions are MPI versions that should run properly ESI Group's software.

Platform	MPI	Interconnect
linux-libc217-x86-64	• Intel MPI 2019	• Ethernet • Mellanox Infiniband • Intel Omni-Path
windows-msvc142 -x86-64	• Intel MPI 2019	• Ethernet • Mellanox Infiniband • Intel Omni-Path

- ***If you are using Intel MPI Library 2018 or older with newer Linux distributions, check out the Troubleshooting chapter.***
- ***If you get the "`Python error: <stdin> is a directory, cannot continue`" error message, check out the Troubleshooting chapter.***

Software Installation on Windows

Installation with the Microsoft Windows InstallShield Wizard

Proceed as follows to launch the software installation:

1. Insert the DVD into the disc drive.
2. Follow the procedure described below, depending on if **autorun** is enabled on your computer:
    - If **autorun** is enabled, the installation should start automatically.
    - If **autorun** is disabled:
        1. Open a Windows Explorer window.
        2. Click your DVD drive icon.
        3. Double- click `VPSolution2023r0_ Windows\Setup.exe` to start the software installation.
3. Follow the instructions displayed in the **InstallShield Wizard** dialog box.
   The installer will successively install each of the components. If one of the component is already installed, it directly switches to the next. You can also select the components you wish to install, however you will need them all to properly run VIRTUAL PERFORMANCE SOLUTION 2023.
4. When the software installation is completed, VIRTUAL PERFORMANCE SOLUTION 2023 is available at the following locations:
    - in the **Start** > **Programs** > **ESI Group** menu;
    - from the **ESI Group** icon folder on the desktop.

Installation with Command Lines

Installation is powered by Microsoft® Windows® Installer. Therefore, installation can be done graphically or silently using command lines:

1. Open a command prompt (`cmd.exe` or `command.com`).
2. Use the following syntax:

```
setup.exe <setup-parameters> /V"<msiexec-parameters> PROPERTY=value
OTHERPROPERTY=^"value with blank space^""
```

   - Obtain list of <setup-parameters> by typing: setup.exe /?
   - Obtain list of <msiexec-parameters> by typing: msiexec /?
   - The most commonly used <msiexec-parameters> that are available:

/I	Installs or configures VIRTUAL PERFORMANCE SOLUTION 2023.
/q [n|b|r|f]	Sets the User Interface (UI) level as follows: ▪ **/q**, **/qn**: No UI ▪ **/qb**: Basic UI. Use **qb!** to hide the **Cancel** button. ▪ **/qr**: Reduced UI. No modal dialog box is displayed at the end of the installation. ▪ **/qf**: Full UI. Any authored **FatalError**, **UserExit**, or **Exit** modal dialog boxes are displayed at the end. ▪ **/qn+**: No UI, except a modal dialog box is displayed at the end. ▪ **/qb+**: Basic UI with a modal dialog box displayed at the end. The modal box is not displayed if you cancel the installation. Use **/qb+!** or **/qb!+** to hide the **Cancel** button. ▪ **/qb-**: Basic UI with no modal dialog box. **/qb+-** is not a supported at UI level. Use **/qb-!** or **/qb!-** to hide the **Cancel** button.

The ! option is available with Microsoft® Windows® Installer 2.0 and works only with basic UI. It is not valid with full UI.

   - The following **PROPERTY** (case sensitive) options are available:

INSTALLDIR="<Path>"	Installation directory. Default value is `%ProgramFiles%\ESI Group\Virtual-Performance\2023.0`

FLEX=FALSE	Specify this line if you do not want to install the FlexNet licensing toolkit. Default value is `TRUE` in MSI table.
VPS=FALSE	Specify this line if you do not want to install VIRTUAL PERFORMANCE SOLUTION 64-bit solvers. Default value is `TRUE` in MSI table.
DOC=FALSE	Specify this line if you do not want to install Documentation files. Default value is `TRUE` in MSI table.
SAMPLES=FALSE	Specify this line if you do not want to install Example files. Default value is `TRUE` in MSI table.

Examples:

The following command line installs silently (**/q**) VIRTUAL PERFORMANCE SOLUTION 2023 in the `D:\Program Files\ESI Group\VPS2023` directory, without example files:

```
msiexec /i "<DriveLetter>:\VPSolution2023r0_Windows\ESI Group Virtual
Performance Solution 2023.0.msi" /q INSTALLDIR="D:\Program Files\ESI
Group\VPS2023\" SAMPLE=FALSE
```

3. When the software installation is completed, VIRTUAL PERFORMANCE SOLUTION 2023 is available at the following locations:
   - in the **Start** > **Programs** > **ESI Group** menu;
   - from the **ESI Group** icon folder on the desktop.

Installing Licenses

Getting your License(s)

Prerequisites

Before starting the license installation you should obtain a valid license file.

VIRTUAL PERFORMANCE SOLUTION 2023 software is version-protected and compiled with Flexera™ **version 11.18.2**.

You must have installed a license server running the same Flexera™ version or higher to be able to run the software.

License files delivered for the previous versions may not be valid with this version of the software and need to be updated to contain VIRTUAL PERFORMANCE SOLUTION 2023 features.

Retrieving Necessary System Information

System information must be retrieved before you can request your Flexera™ license files. Proceed as follows to do so:

1. For each PC machine that will be running VIRTUAL PERFORMANCE SOLUTION 2023, execute the provided shortcut **Quick Access to my Station HostID** or the corresponding command line:

   `c:\flexlm\ESIGetHostID.bat`.

   This utility creates a text file on the user's desktop, which collects all required details (**Operating System**, **Hostname**, **Hostid settings**) to be transferred to your local ESI support representative, who will provide you with the corresponding license file.

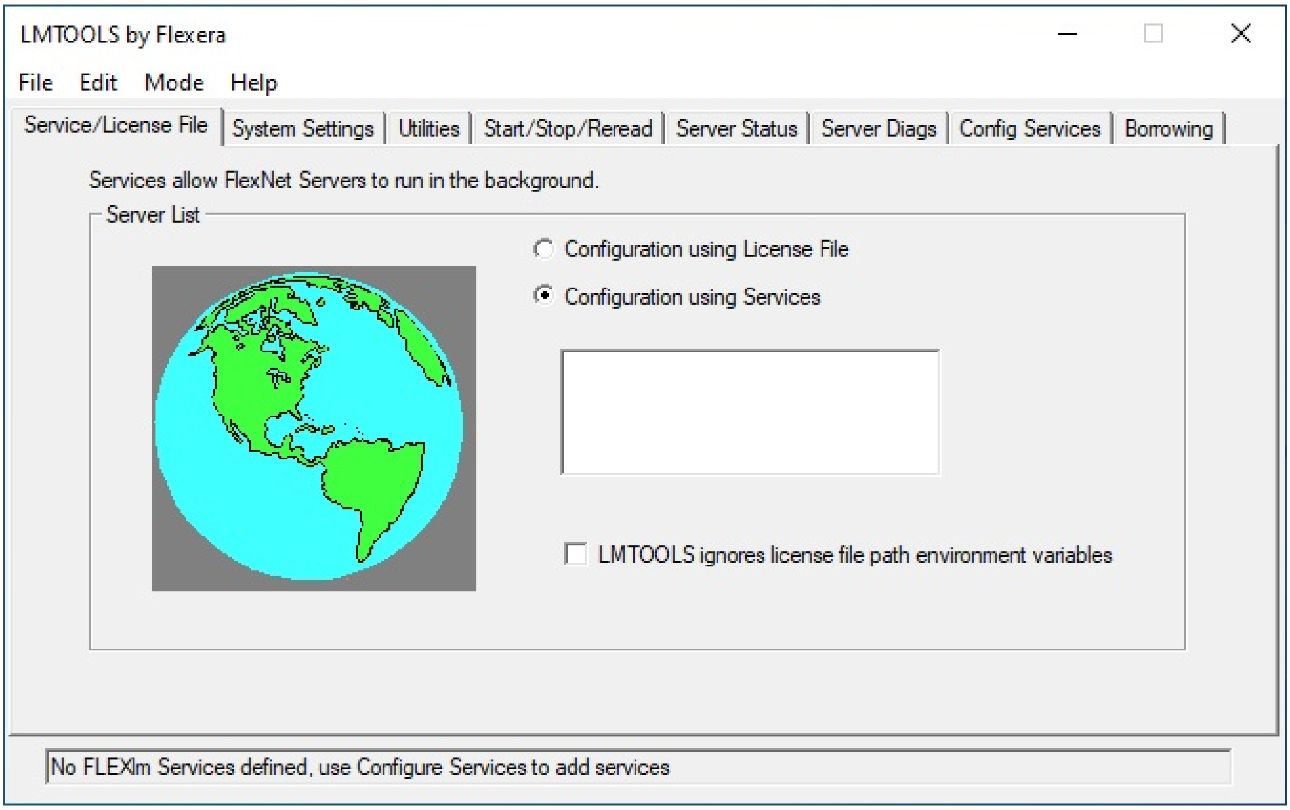
2. Once you have received your license file(s), move it(them) to the appropriate `c:\flexlm` directory on each machine with the `pam_lmd.lic` filename.

Starting Flexera™ License Server

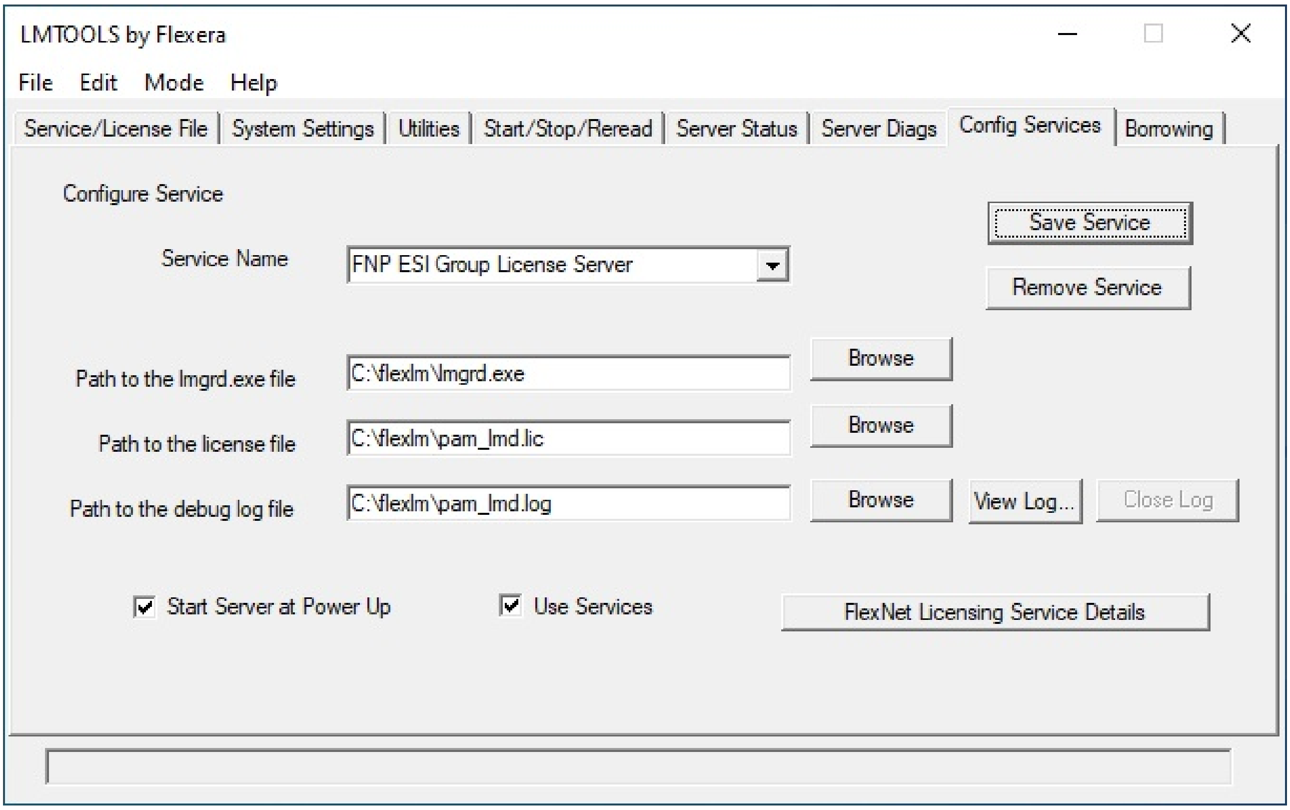
The Flexera™ license server installation must be done with administrator privileges.

Once the VIRTUAL PERFORMANCE SOLUTION 2023 installation is completed, proceed as follows:

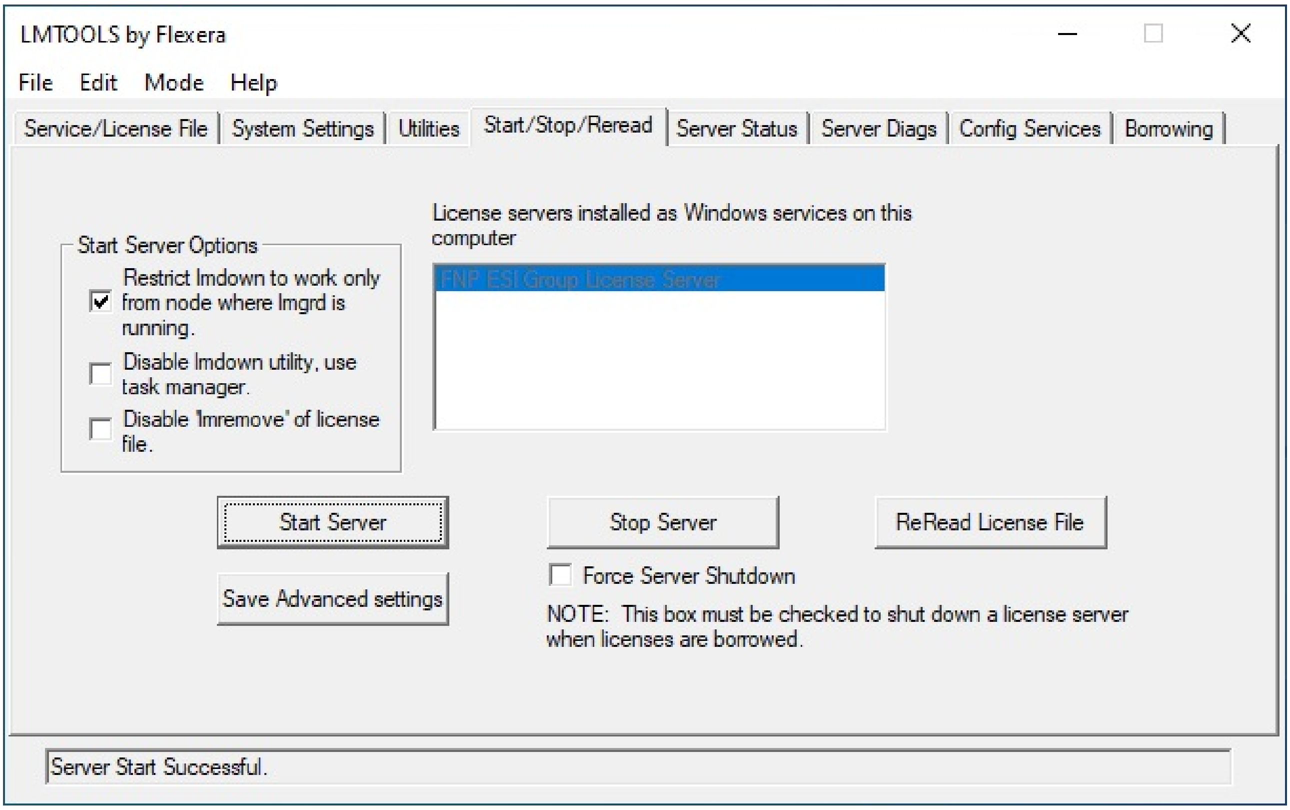
1. Move the `pam_lmd.lic` file to the `c:\flexlm` directory (if not already done).
2. Edit the `pam_lmd.lic` file using any usual text editor to change `$PAMHOME` variable to `c:\flexlm\pam_lmd.exe` on the line beginning with the **`VENDOR`** keyword.
3. Start the license server:
   a. Execute **`c:\flexlm\lmtools.exe`**.
   b. Go to the **Service/License File** tab and select **Configuration using Services**.



c. Select the **Config Services** tab, adjust parameters as shown below and click the **Save Service** button.



d. Select the **Start/Stop/Reread** tab and click the **Start Server** button.



After completing these installation steps, you may start using VIRTUAL PERFORMANCE SOLUTION 2023.

Additional specific Flexera™ documentation is provided with the software in the `c:\flexlm` *directory.*

Software Installation on Linux

Overview

The following installation procedures are available to install VIRTUAL PERFORMANCE SOLUTION 2023 on Linux platforms:

- using the Graphical User Interface script `paminst-gui.sh`;
- using the command line script **`paminst.sh`**.

> ***Sufficient privileges may be required by the system during the installation of ESI Group products. Although no system library is updated in the process, sufficient privileges are necessary to mount the DVD and install the boot files to start the license server.***

Both procedures consists of the following tasks:

1. Installing the software: all software files are copied from the installer to the chosen location.
2. Setting the license file:
    a. Installing the license file.
    b. Starting the license daemons `lmgrd` and `pam_lmd` if needed.
    c. Installing the license daemon startup scripts.
3. Customizing the installation by setting up the ESI Group software environment for each current user.

Tasks 2 and 3 can be performed later if you do not want to do them immediately.

After installation, files installed on the hard disk are organized in a directory tree as shown below.

HEAD DIRECTORY
Referenced as `PAMHOME` installation scripts, ...

- **MODULE1**
  Executables, other files, ...
  - **Version1_COMMON**
    Resource files
  - **Version1_platformOS1**
    Executables
  - **Version1_platformOS2**
    Executables
  - **Version2_platformOS1**
    Executables
- **EXAMPLES**
  Example files, testcases, ...
- **LICENSES**
  License files
- **FLEXnet**
  Daemon and tools
  - **platformOS1**
    Daemon and tool files
  - **platformOS2**
    Daemon and tool files
- **DOC**
  Electronic documentation

DVD Mounting Instructions

Proceed with the following steps for DVD mounting:

1. Insert the DVD into the drive.
2. Set the **`exec`** mount option (**`eject`** to dismount).
3. When the installation is done, quit the `/cdrom` directory before ejecting the media from the drive.

Installation with the ESI Installation Wizard

System Requirements

The following is necessary to install the software properly:

- An X-compliant window system available on your display.
  Position the `DISPLAY` environment variable for display on the current screen and, if necessary, allow this display using an appropriate **`xhost`** command on the display host.
- About 15 Mb of free space in the `/tmp` directory to store graphical tools temporarily.

Launching Installation

Proceed as follows:

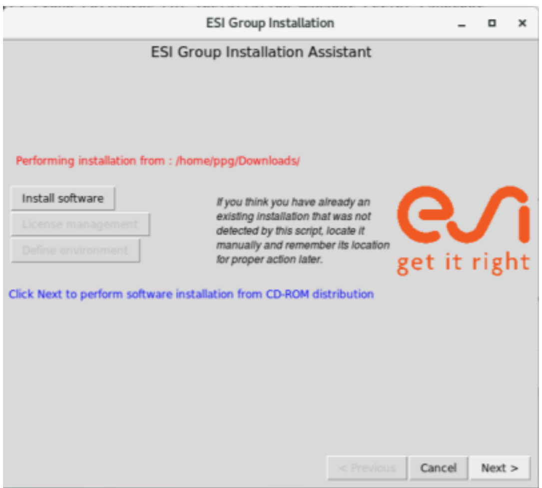
1. Type one of the following commands from the `cdrom` root directory:

   ```
   ./INSTALL.SH
   ```

   or

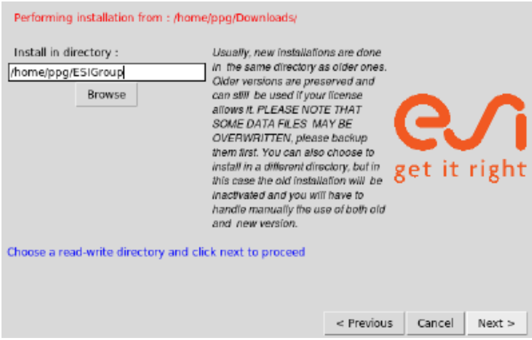
   ```
   /<drive_mount_point>/INSTALL.SH
   ```

   If this error message is obtained: `INSTALL.SH: /bin/sh: bad interpreter: Permission denied`, refer to "Linux: How to Override the DVD/CD-ROM Drive Protection?" for further details.
2. After the graphical tools are loaded in `/tmp`, the main dialog box window appears.



3. Click the **Install software** button and click the **Next** button to continue.
4. Specify the installation directory in the **Install in directory** text field. By default, it is filled with the path name used with the previous ESI Group software installation, if any. To change it, proceed with one of the following methods:

- Fill in the text field with a valid path name directory (read, write, execute permissions are required), or
- Click the **Browse** button to select a software installation location different from the default one.



5. Click the **Next** button when you are satisfied with the software installation directory selected.

Selecting Components to be Installed

1. Select all packages you wish to install on the system by checking the corresponding boxes:
   - **flexnet_v11.18.2**: FlexNet v11.18.2 utilities for Linux platforms.

     This package is required to start a license server when a floating license file is delivered. It is not required for a demo or uncounted node-locked license files.
   - **VisualEnv-2023.0_Linux64**: Visual-Environment 2023.0 for Linux x86 64-bit processors
   - **VPS-Solvers_Linux-Intel64**: VIRTUAL PERFORMANCE SOLUTION 2023 DMP solvers for Linux, optimized for Intel Xeon processors
   - **examples**: Explicit and Implicit example files for VIRTUAL PERFORMANCE SOLUTION 2023
2. Click the **Next** button to continue. The 'file downloading' window is displayed.
3. When downloading is finished, click the **Close** button in the bottom of the window to continue. Installation of files is now completed.
4. In the displayed window:
   - Click the **Exit** button if you wish to perform the license installation and software customization later.
   - Click the **License management** button to install the license. A license file is needed to perform this task. For more details, refer to “Installing Licenses”.
   - Click the **Define environment** button to customize the software environment. For more details, refer to “Customizing Installation”.

If you encounter some issues with executing scripts from the DVD drive, refer to “Linux: How to Override the DVD/CD-ROM Drive Protection?” for help.

Installing Licenses

System Requirements

Before starting the license installation you should obtain a valid license file.

VIRTUAL PERFORMANCE SOLUTION 2023 software is version-protected and compiled with Flexera™ **version 11.18.2**.

You must have installed a license server running the same Flexera™ version or higher to be able to run the software.

License files delivered for the previous versions may not be valid with this version of the software and need to be updated to contain VIRTUAL PERFORMANCE SOLUTION 2023 features.

The license installation should be performed on the system where the license server will run.

Launching License Installation

To launch the license installation with the ESI's Installation Wizard, proceed as follows:

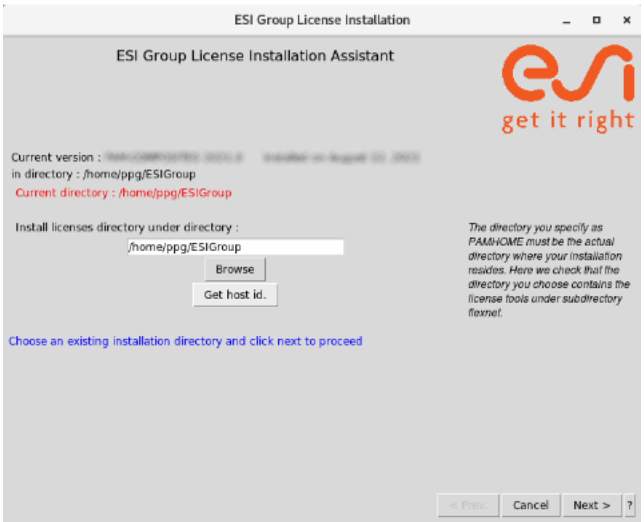
1. Go to the software installation directory and type one of the following commands:

   ```
   ./paminst-gui.sh  lic
   ```

   or

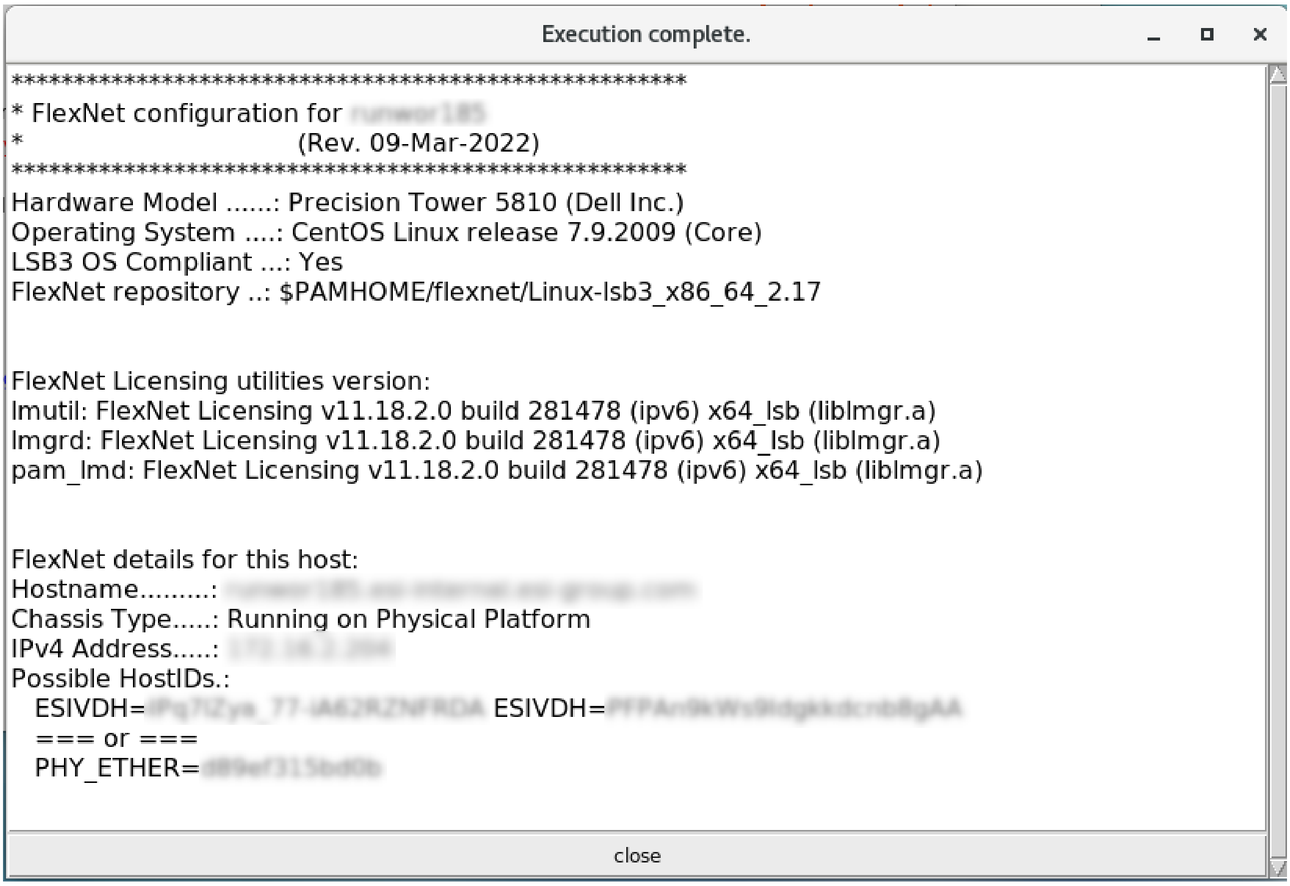
   ```
   ./paminst-gui.sh  install_lic
   ```

2. After the graphical tools are loaded in `/tmp`, the ESI Installation Wizard dialog box appears.



Getting the Host ID

Click the **Get host id.** button to get the system information (**hostname**, **hostid**) needed for the license server (floating license) or standalone license (node-locked license) installation. The following window should be displayed:



<u>Important remark on Linux i686 or x86_64 based distributions</u>

The FlexNet Toolkit can only run on LSB3™ (Linux Standard Base™)-compliant Linux distributions.

The 'LSB3 OS Compliant' detail must display 'Yes', otherwise your Linux distribution cannot run a license server with this latest version of the FlexNet Toolkit.

Check that the following package is installed on the station:

- ***RedHat Based Operating Systems:***
  - ***For RedHat, CentOS, Oracle Linux, Fedora ...: package*** `redhat-lsb`
  - ***How to check installation:*** `rpm -q redhat-lsb`
  - ***How to install it (as root or with sudo):*** `yum install redhat-lsb`
- ***SUSE Based Operating Systems:***
  - ***For SUSE Entreprise, openSUSE ...: package*** `lsb`
  - ***How to check installation:*** `rpm -q lsb`
  - ***How to install it (as root or with sudo):*** `zypper install lsb`
- ***Debian Based Operating Systems:***
  - ***For Debian, Ubuntu ...: package*** `lsb`
  - ***How to check installation:*** `dpkg -l lsb`
  - ***How to install it (as root or with sudo):*** `apt install lsb`

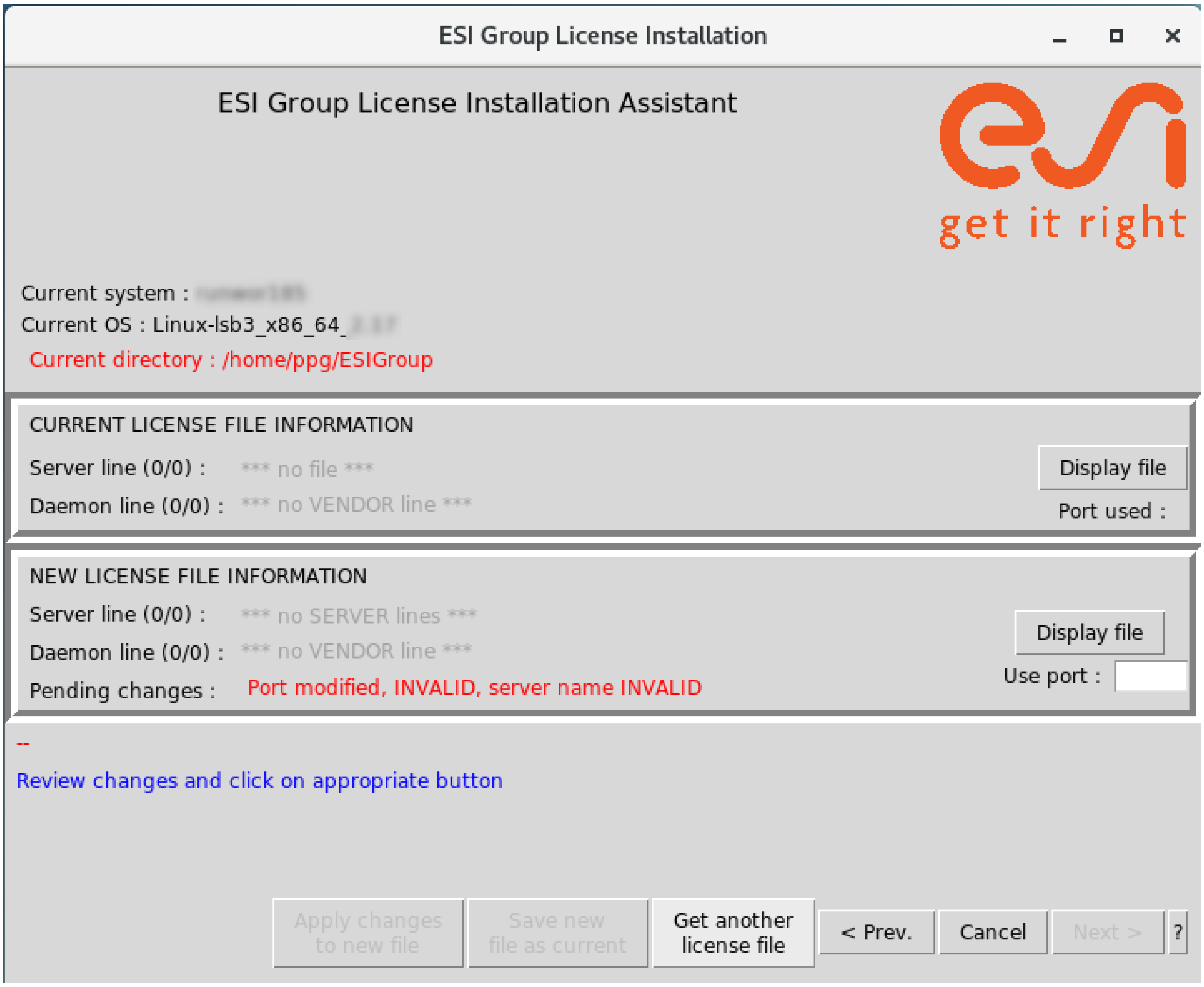
Selecting License Files and License Installation Directory

By default, the **Install licenses directory under directory** text field is filled with the path name of the previous ESI Group's software installation, if any.

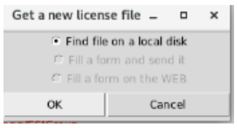
1. Click the **Browse** button to redefine the path name of the software installation directory.

   The directory specified in the text field should contain a `licenses` *sub-directory with write permission for the* ***Next*** *button to be available.*

2. Click the **Next** button when the installation directory path is selected.
   The license file is installed and the following window is displayed:



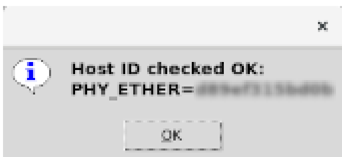
3. Click the **Get another license file** button to select a new license file.
4. Select **Find file on a local disk** in the displayed window and click **OK**.



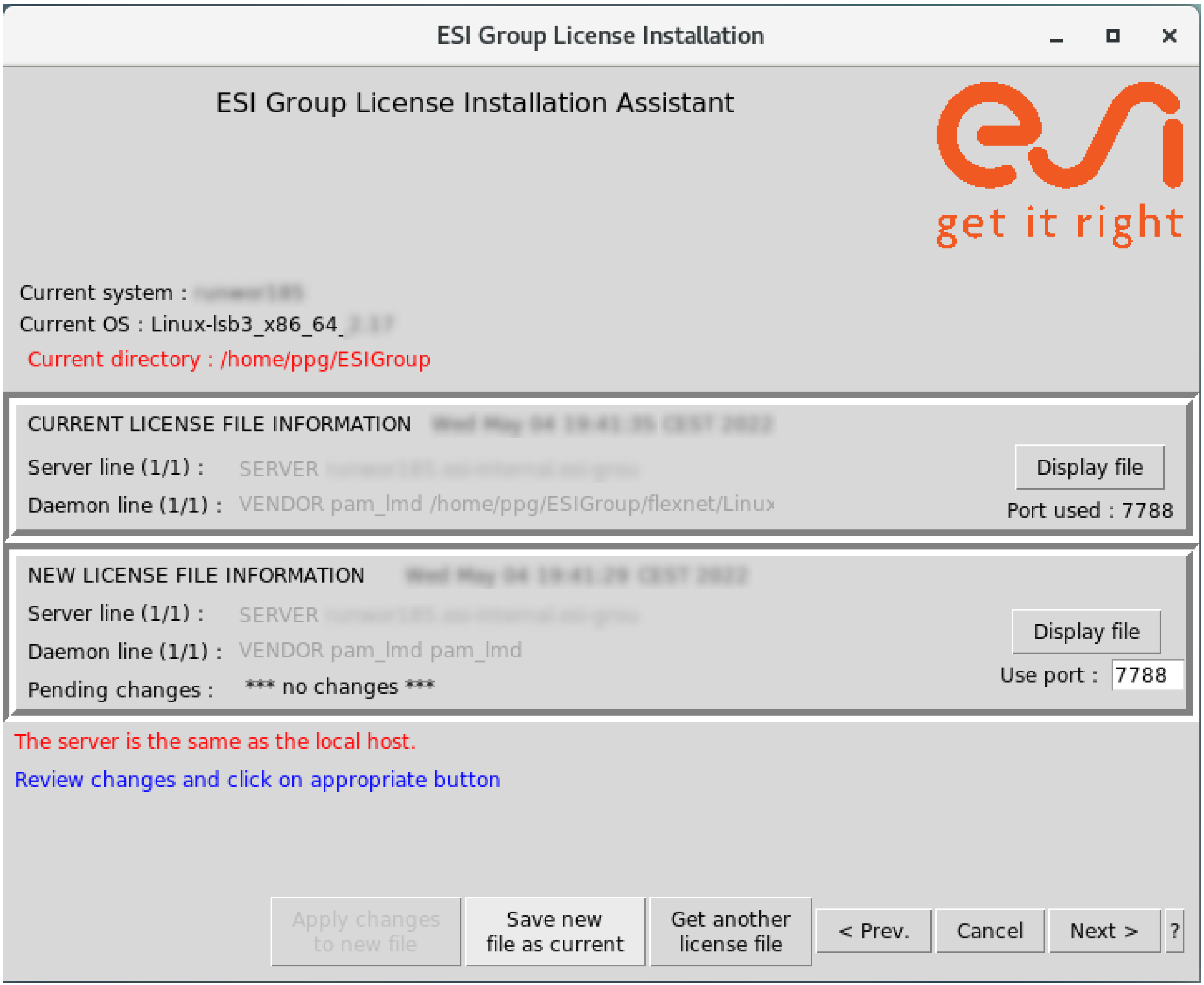
5. Select the new license file and click the **Open** button to validate the selection.

By default, the file filter is set to find PAM license file format.

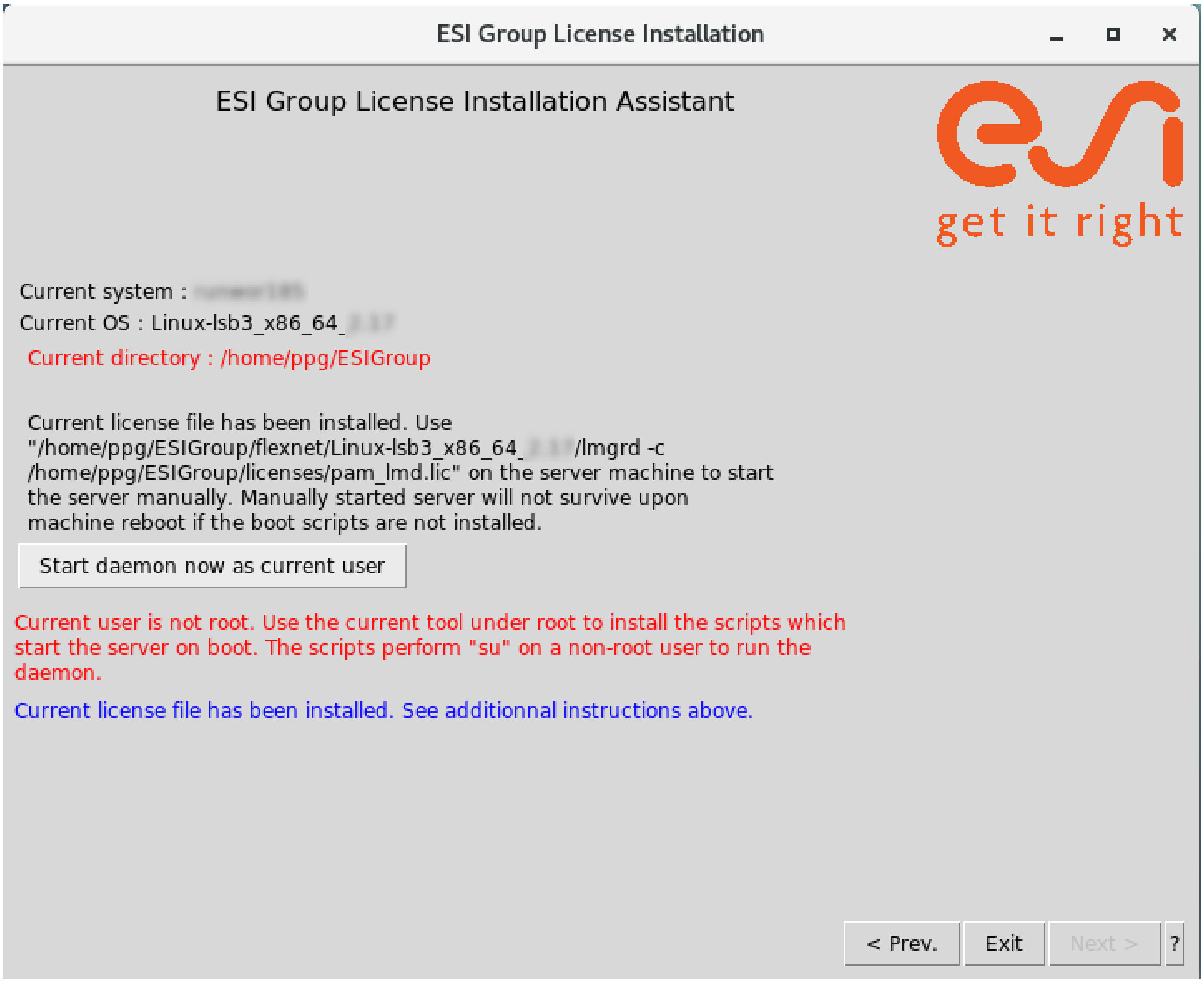
A small window pops up displaying the checking result of the selected license file with the Host ID.



The dialog box of license file selection will be displayed again and filled in with information corresponding to the selected license file:



6. Click the **Apply changes to new file** button and then the **Save new file as current** button.
   The new license file is installed.
7. Click the **Next** button to start the license daemons.
8. Click the **Start daemon now as current user** button to start the daemon.



Additional specific Flexera™ FlexNet documentation is provided with the software in the installation directory.

Customizing Installation

The ESI Group software installation script `paminst.sh` can be used for customization.

The customization script creates definitions to allow access to the executables and configuration files for all users or for selected users, depending on the method selected.

The software configuration is defined in the following files:

- `psi.Cenv` for csh and derivatives;
- `psi.Kenv` for ksh and derivatives;
- `psi.Baenv` for bash.

> ***sh*** *is not supported due to its lack of functionalities necessary for ESI Group products.*
>
> *We recommend to* ***sh*** *users to migrate to* ***ksh*** *which preserves* ***sh*** *syntax and habits.*

This customization is done with `pamcust.sh` which can be used separately as well.

To perform the customization separately, proceed as follows:

1. Log in onto the machine on which you want to perform the customizations.
2. Change `$PAMHOME` variable to the directory where the installation was done.
3. Type **`./pamcust.sh`** and follow the instructions.

   The customization procedure will automatically update the `.cshrc`, `.profile` or `.bash_profile` files accordingly, as shown below:

C Shell – `.cshrc`	`### ESI-Group begin ###`
	`#`
	`#ESI-Group Software environment`
	`#setenv PAMHOME /home/ppg/ESIGroup`
	` setenv PAMENV $PAMHOME/env-`` `uname` ``
	`   if ( -r $PAMENV/psi.Cenv ) then`
	`     source $PAMENV/psi.Cenv`
	`   endif`
	`### ESI-Group end ###`

Korn Shell – `.profile`	<pre>### ESI-Group begin ###
	#
	#ESI-Group Software environment#
	 PAMHOME=/home/ppg/ESIGroup; export PAMHOME
	 PAMENV=$PAMHOME/env-\`uname\`; export PAMENV
	 if [-r $PAMENV/psi.Kenv]; then
	 . $PAMENV/psi.Kenv
	 fi
	 ENV=${ENV:””}
	 if [-z “$ENV”]; then
	  ENV=$HOME/.kshrc; export ENV
	 fi
	### ESI-Group end ###</pre>

Bourne Again Shell – `.bash_profile`	<pre>### ESI-Group begin ###
	#
	#ESI-Group Software environment#
	 PAMHOME=/home/ppg/ESIGroup; export PAMHOME
	 PAMENV=$PAMHOME/env-\`uname\`; export PAMENV
	 if [-r $PAMENV/psi.Baenv]; then
	  . $PAMENV/psi.Baenv
	 fi
	 if [-z “$BASH_ENV”]; then
	  BASH_ENV=$HOME/.bashrc; export BASH_ENV
	 fi
	### ESI-Group end ###</pre>

4. Repeat this procedure for each candidate user.

Troubleshooting

Windows: What to Do if MAC Address is "FFFFFFFF"?

Issue Description

The licensing tools see my **hostid** as "FFFFFFFF" when using a Virtual NIC even though the **`ipconfig`** command shows a different **hostid** for the Virtual NIC.

Workaround

Virtual NICs is no longer supported since FlexNet 11.6.

This is not a bug, but a deliberate change in FlexNet software behavior.

Workaround is to un-team these Ethernet adapters or enable a disabled adapter or install a new Ethernet adapter.

If not possible, "Disk Volume Serial Number" can eventually be used as server **hostid**.

Linux: Resolving Segmentation Fault in Legacy Intel MPI Library on Newer Linux Distributions

Issue Description

Attempting to use Intel MPI Library 2018 or older on newer Linux operating systems will lead to segmentation faults. The segmentation fault is due to an incompatibility in `glibc`, and will show in a callstack as:

```
1) libc.so.6 0x0008a5b6 ! strtok_r :?
```

Resolution

Update to the latest version of the Intel MPI Library. If it is not possible, check out the "Workaround" below.

Workaround

If you must remain on an older version of the Intel MPI Library, proceed as follows:

1. Create a `strtok_proxy.c` file with the following code:

```
#define _GNU_SOURCE 1

#include <string.h>
#include <dlfcn.h>

typedef char *(*strtok_t) (char *str, const char *delimiters);
static strtok_t strtok_orig_p = NULL;
char *strtok(char *str, const char *delimiters)
{
 if (strtok_orig_p == NULL) {
   strtok_orig_p = (strtok_t) dlsym(RTLD_NEXT, "strtok");
   if (strtok_orig_p == NULL) {
     /* Error handling */
     return NULL;
   }
   if (str == NULL) {
     str = "";
   }
}

 return strtok_orig_p(str, delimiters);

}
```

2. Compile this file using the following command:

```
$ gcc -c -Wall -Werror -fpic ./strtok_proxy.c
$ gcc -ldl -shared -o ./strtok_proxy.so ./strtok_proxy.o
```

3. Apply the generated library at runtime using the following command:

```
$ export LD_PRELOAD=./strtok_proxy.so
```

Linux: "`Python error: <stdin> is a directory, cannot continue`"

Issue Description

You may get the following error message when you are running with Intel MPI Library 2019 or later on AMD:

```
Python error: <stdin> is a directory, cannot continue
```

Resolution

Use Intel MPI Library 2018.3 instead, or if not possible, use the **`"-s all"`** argument of MPI:

```
pamcsm -mpiext "-s all"
```

Linux: How to Override the DVD/CD-ROM Drive Protection?

Issue Description

In some Linux distributions, DVD/CD-ROM drives may be protected to not run malicious code. This means that users do not have the right to execute scripts directly from the drive.

The following error message may appear:

```
/media/<product>/INSTALL.SH: /bin/sh: bad interpreter: Permission denied
```

It means the drive is protected and has the **`noexec`** mount option.

Workaround

The script `INSTALL.SH` can override this protection.

You can follow these instructions to get around:

1. Copy the `INSTALL.SH` script to a temporary directory.
2. Execute it by giving the drive mount point as argument.

Example

Let us consider that `/media/<product>` is the drive mount point:

```
mkdir install_temp
   cd install_temp
     cp /media/<product>/INSTALL.SH .
     ./INSTALL.SH /media/<product>

     #################################################################
     ESI Group Softwares GUI Installation Manager Script Launcher
     Copyright 1998 ~ 2023   ESI GROUP
     #################################################################
     Creating temporary installer directory... done.
     --------------------------------------------------------------------
     Invoking installation of <product> <version>
     --------------------------------------------------------------------
     ESI Group GUI INSTALLATION MANAGER SCRIPT
     Copyright 1998~2023,     ESI GROUP
     This script manages the gui scripts.

     To start the ESI Group software installation (from the CD only)
     Type    './Paminst-gui.sh'
     or      './Paminst-gui.sh paminst'

     To start the license management (from the target directory only)
```

```
Type    './Paminst-gui.sh lic'
or      './Paminst-gui.sh install_lic' still accepted for
compatibility.

---------------------------------------------------------------------
Unloading installation tools, 15 Mb are necessary in /tmp,
please wait...
Tools ready.
```

Only scripts files are copied in this temporary installer directory; archives containing the files to install are on the media.

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